



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF AGRICULTURAL ENGINEERING
B.E. Agricultural Engineering
CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2	Exempted for LE	HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3	Exempted for LE	HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
4	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
5	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
6		HS	AG0801	NA	Entrepreneurship Development and Business Management	-	2	0	2	CGPA	Theory
7	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
8	Exempted for LE	BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
9		BS	SH2224	19MA224	Transforms and Differential Equations	-	4	0	4	CGPA	Theory
10	Exempted for LE	BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
11	Exempted for LE	BS	SH3203	19PH203	Engineering Physics	-	2	1	3	CGPA	TIP (6 Expts)
12	Exempted for LE	BS	SH4201	19CY201	Engineering Chemistry	-	2	1	3	CGPA	TIP (6 Expts)
13	Exempted for LE	BS	AG0802	NA	Principles of Soil Science	-	2	0	2	CGPA	Theory
14	Exempted for LE	BS	AG0803	NA	Soil Science Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
15	Exempted for LE	ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
16		ES	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
17	Exempted for LE	ES	ME0309	19ME309	Engineering Drawing and Visualisation	-	3	1	4	CGPA	TIP (6 Expts)
18		ES	ME0310	19ME310	Engineering Practices Lab	-	0	1	1	CGPA	Practical (6 Expts)
19	Exempted for LE	ES	ME0801	NA	Theory of Machines	-	3	1	4	CGPA	TIP (6 Expts)
20	Exempted for LE	ES	EC2306	19EE306	Basic Electrical, Electronics and Instrumentation Engineering	-	2	1	3	CGPA	TIP (6 Expts)
21		ES	AG0807	NA	Fluid Mechanics and Open Channel Hydraulics	-	2	0	2	CGPA	Theory
22		ES	AG0808	NA	Fluid Mechanics Laboratory	-	0	1	1	CGPA	Practical (6 Expts)

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
23		ES	AG0806	NA	Soil Mechanics	-	1	1	2	CGPA	Theory (Int Only) and Practical (6 Expts)
24		ES	AG0809	NA	Thermodynamics, Refrigeration and Air Conditioning	-	3	0	3	CGPA	Theory
25	Exempted for LE	ES	AG0804	NA	Surveying and Levelling	-	2	0	2	CGPA	Theory
26	Exempted for LE	ES	AG0805	NA	Surveying Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
27		ES	ME0311	19ME311	Engineering Mechanics	-	4	0	4	CGPA	Theory
28		ES	ME0312	19ME312	Mechanics of Solids	-	3	1	4	CGPA	TIP (6 Expts)
29		ES	ME0407	19ME407	Design of Machine Elements	19ME312 / ME0312 / 19ME304	3	1	4	CGPA	TIP (6 Expts)
30		ES	CS3410	19AI410	Introduction to Machine Learning	-	2	1	3	CGPA	TIP (6 Expts)
31	Exempted for LE	PC	AG0810	NA	Principles and Practices of Crop Production	-	2	0	2	CGPA	Theory
32	Exempted for LE	PC	AG0811	NA	Crop Production Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
33		PC	AG0812	NA	Watershed Hydrology	-	2	0	2	CGPA	Theory
34		PC	AG0813	NA	Watershed Hydrology Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
35		PC	AG0814	NA	Unit Operations in Agricultural Processing	-	2	0	2	CGPA	Theory
36		PC	AG0815	NA	Unit Operations Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
37		PC	AG0816	NA	Irrigation and Drainage Engineering	-	3	0	3	CGPA	Theory
38		PC	AG0817	NA	Irrigation Field Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
39		PC	AG0818	NA	Tractor and Farm Engines	-	2	0	2	CGPA	Theory
40		PC	AG0819	NA	Agricultural Structures and Cost Estimation	-	3	0	3	CGPA	Theory
41		PC	AG0820	NA	Engineering Properties of Agricultural Produce	-	2	0	2	CGPA	Theory
42		PC	AG0821	NA	Renewable Energy Sources for Agricultural Engineering	-	3	0	3	CGPA	Theory
43		PC	AG0822	NA	Renewable Energy Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
44		PC	AG0823	NA	Micro Irrigation Systems and Automation	AG0816	2	1	3	CGPA	TIP (6 Expts)
45		PC	AG0824	NA	Bioenergy Systems and Applications	AG0821	2	0	2	CGPA	Theory
46		PC	AG0825	NA	Post Harvest Engineering of Agricultural Produce	-	3	0	3	CGPA	Theory
47		PC	AG0826	NA	Post Harvest Engineering Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
48		PC	AG0827	NA	Tractor Systems and Controls	AG0818	2	0	2	CGPA	Theory

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
49		PC	AG0828	NA	Tractor and Farm Engines Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
50		PC	AG0829	NA	Farm Machinery and Equipment I	-	2	0	2	CGPA	Theory
51		PC	AG0830	NA	Soil and Water Conservation Engineering	AG0812	3	0	3	CGPA	Theory
52		PC	AG0831	NA	Groundwater Hydrology and Well Hydraulics	-	3	0	3	CGPA	Theory
53		PC	AG0832	NA	Remote Sensing and GIS Applications	-	2	0	2	CGPA	Theory
54		PC	AG0833	NA	Remote Sensing and GIS Applications Laboratory	-	0	1	1	CGPA	Practical (6 Expts)
55		PC	AG0834	NA	Farm Machinery and Equipment II	AG0829	2	0	2	CGPA	Theory
56		PC	AG0835	19AG419	Tractors and Farm Machinery Operation and Maintenance	19AG412/19 AG418	0	2	2	CGPA	Practical (10 Expts)
57		PC	AG0836	NA	Dairy and Food Engineering	-	3	0	3	CGPA	Theory
58		PE(Honours 1)	AG0837	NA	Farm Power and Machinery Management	-	3	0	3	CGPA	Theory
59		PE(Honours 1)	AG0838	NA	Automation and Robotics in Farm Machinery	-	3	0	3	CGPA	Theory
60		PE(Honours 1)	AG0839	NA	Human Engineering and Safety in Farm Machinery Operations	-	3	0	3	CGPA	Theory
61		PE(Honours 2)	AG0840	NA	Watershed Planning and Management	-	3	0	3	CGPA	Theory
62		PE(Honours 2)	AG0841	NA	Irrigation Water Quality and Waste Water Management	-	3	0	3	CGPA	Theory
63		PE(Honours 2)	AG0842	NA	Precision Farming Technologies	-	3	0	3	CGPA	Theory
64		PE(Honours 3)	AG0843	NA	Development of Processed Food Products	-	3	0	3	CGPA	Theory
65		PE(Honours 3)	AG0844	NA	Storage and Food Packaging Technology	-	3	0	3	CGPA	Theory
66		PE(Honours 3)	AG0845	NA	Food Quality Assurance and Management	-	3	0	3	CGPA	Theory
67		PE(Honours 3)	AG0846	NA	Food Plant Design	-	3	0	3	CGPA	Theory
68		PE(Honours 1)	AG0520	19AG520	Agricultural Waste Management	-	3	0	3	CGPA	Theory
69		PE(Honours 2)	AG0521	19AG521	Climate Change and Adaptation	-	3	0	3	CGPA	Theory
70		PE(Honours 2)	AG0522	19AG522	Data Science for Soil and Water Management	-	3	0	3	CGPA	Theory
71		PE(Honours 1)	AG0523	19AG523	Computational Software for Crop Growth Modeling	-	3	0	3	CGPA	Theory
72		PE(Honours 1)	AG0519	19AG519	Solar Energy in Agricultural Systems	-	3	0	3	CGPA	Theory
73		PE(Honours 1)	AG0518	19AG518	Testing and Evaluation of Tractors and Farm Machinery	-	3	0	3	CGPA	Theory
74		PE(Honours 3)	AG0524	19AG524	Emerging Technologies in Food Processing	-	3	0	3	CGPA	Theory
75		PE(Honours 3)	AG0525	19AG525	3D Printing Technology for Food Industry	-	3	0	3	CGPA	Theory

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
76		OE	AG0854	NA	Principles of Agricultural Engineering	-	3	0	3	CGPA	Theory
77		OE	AG0855	NA	Urban Agriculture	-	3	0	3	CGPA	Theory
78		OE	AG0856	NA	Process Engineering of Fruits and Vegetables	-	3	0	3	CGPA	Theory
79		EEC	AG0849	NA	Directed Studies	-	0	2	2	CGPA	Practical (10 Expts)
80		EEC	AG0707	19AG707	Educational Tour	-	0	2	2	Non - CGPA	Practical (10 Expts)
81		EEC	AG0850	NA	Experiential Learning on Campus	-	0	3	3	CGPA	Practical (10 Expts)
82		EEC	AG0852	NA	Industrial Training	-	0	6	6	CGPA	Project
83	Option 1	EEC	AG0851	19AG701	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
84	Option 1	EEC	AG0853	NA	Project Work Phase II	-	0	6	6	CGPA	Project
85	Option 2		SH6802	NA	Start Up	-	0	8	8	CGPA	Project
86		EEC	AG0848	NA	Skill Development Training	-	0	6	6	CGPA	Project
87	Exempted for LE	EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
88	Exempted for LE	EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
89		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
90		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
91		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
92		EEC	NA	19EY706	Company Specific Assessments For Employability	-	0	1	1	CGPA	Practical (6 Expts)
93		MC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
94		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
95	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
96	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
97	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	12	2
Basic Science	23	4
Engineering Science	45	27
Professional Core	54	54
Professional Electives	9	9
Open Electives	6	6

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
					Employability Enhancement Courses	33	33				
					Mandatory Courses	2	2				
					TOTAL	184	137				